SEMESTER 5 · FINAL REPORT · TMH OSTEONECROSIS AI PROJECT

Predicting Osteoradionecrosis of the Jaw from Dose and Clinical Data, and a Study of CT-based Jaw Sub-volume Segmentation (IMPACT)

Part A builds and evaluates a machine-learning pipeline that predicts the grade of osteoradionecrosis (ORN) for 1,259 head and neck cancer patients. Part B studies the IMPACT paper from the MD Anderson Head and Neck group, explores its public CT data, audits its code and plans how to reproduce it.

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Date	November 2026
Code	github.com/MuaazSM/orn-pipeline (Part A); impact-seg/ (Part B)
Status	Research study. Not validated for clinical use.

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Abstract

Osteoradionecrosis (ORN) of the jaw is a late, painful complication of radiotherapy for head and neck cancer. Published prediction models answer a yes or no question. We asked whether the same planning data, the dose delivered to the mandible and routine clinical factors, can also predict how severe the ORN will be.

Part A. We built a reproducible pipeline (scikit-learn, XGBoost, LightGBM, SHAP, FastAPI and a Next.js dashboard) on the 1,259-patient MD Anderson dataset released by van Dijk et al. ORN occurred in 173 patients (13.7%). On a held-out test set of 252 patients a random forest predicted whether ORN occurs with a ROC AUC of 0.74 (95% CI 0.66 to 0.81), matching the published benchmark of 0.74. Predicting the grade did not work: macro-F1 was 0.40 for three classes, 0.42 for a two-stage model and 0.21 for the five grades, against chance levels of 0.36, not defined and 0.20. Dose in the mid range of the mandible (D25 to D35, V45 to V55) drove the model, which agrees with earlier work.

Part B. The IMPACT paper (Humbert-Vidan et al., medRxiv 2025) segments 12 mandible and maxilla sub-volumes and 32 teeth on planning CT, so that dose can be read per region. We downloaded its public release (60 CT scans, three label sets and dose maps for 8 patients), explored every scan, audited all five training scripts and estimated the cost of reproducing the work. Maxilla basal bone is present in only 25 to 29 of 60 scans and is 0.1 to 0.8 mL at the median, which explains why the authors could not segment it. The released code does not contain Swin UNETR, the inference stage, the five-fold vote or the metrics. We wrote our own MONAI implementation and tested its plumbing on synthetic data. No model has been trained on the real data.

Link between the parts. Region-level dose from IMPACT-style segmentation could replace the single whole-mandible dose-volume histogram used in Part A. That needs a cohort with both CT and ORN outcomes, which the public IMPACT release does not provide.

1. Introduction

1.1 Clinical background

Radiotherapy for head and neck cancer is highly conformal, but the mandible and maxilla sit close to the target and always receive some dose. Radiation damages the bone's blood supply and cells. Irradiated bone then heals poorly after a dental extraction or other trauma, and in the worst case the bone dies and becomes exposed. This is osteoradionecrosis. Decisions about extractions, implants and dose distribution before, during and after treatment all depend on estimating this risk.

ORN is usually graded 0 to 4 in the dataset used here, from no ORN to the most severe form. Severe grades carry a large burden for the patient, so a tool that tells clinicians not only whether but how badly a patient might be affected would be useful.

1.2 Objectives

1. Build a reproducible pipeline that turns the MD Anderson ORN dataset into graded risk predictions with explanations.
2. Test honestly whether the grade, and not only the presence, of ORN can be predicted from dose and clinical data.

3. Study the IMPACT paper as a route to region-level dose features: what its data and code contain, what our exploration shows, and what it would take to train its models ourselves.
4. Describe how the two parts connect and what the next semester should do.

1.3 What this report contains

The report combines two earlier slide decks. The first ("Predicting ORN Severity") covers Part A. The second ("IMPACT Implementation Study") covers Part B. We rechecked the numbers in both against the saved result files, the data and the code, and list every discrepancy we found in Appendix B. Where this report differs from a slide, the report is the corrected version.

A note on authorship of the sources: the IMPACT paper, its data and its code are the work of the MD Anderson group. The second deck is our own study of that work, not a deck by the authors.

2. Background and related work

Study	Predicts	Data	Best result	Relevance
van Dijk et al. 2021 (NTCP)	ORN yes / no	MD Anderson, 1,259 pts	AUC 0.74 to 0.77	Our dataset and benchmark; D30 and extraction are the key predictors
Reber et al. 2023	ORN yes / no	same cohort	F1 0.31 (logistic regression)	Logistic regression beat a ResNet (F1 0.07), so we use classical models
PREDMORN 2025	ORN yes / no	see original	AUC 0.65 to 0.69	Also finds D30 important (as stated in our slides)
This work (Part A)	ORN grade 0 to 4	same cohort	AUC 0.74 (binary); grade macro-F1 0.21 to 0.42	Extends the question to severity
IMPACT 2025 (Part B)	Jaw segmentation	60 planning CTs	Dice 0.63 to 0.85	Route to region-level dose

Table 1. Related work. Citations for Reber et al. and PREDMORN are as given in our slides; their full references should be confirmed before submission.

The **ClinRad** ORN staging system, endorsed by ASCO, grades the vertical extent of bone damage seen on imaging. It motivates IMPACT's split of each jaw into alveolar bone (around the teeth, spongy, vascular, containing marrow) and basal bone (denser, mostly cortical).

Part A. ORN grade prediction

3. Data and exploration

3.1 The dataset

The data are the anonymised cohort of van Dijk et al. (2021): 1,259 head and neck cancer patients treated at MD Anderson between 2005 and 2015, distributed as one CSV with 54 columns. They hold demographics (age, sex), smoking status and pack-years, tumour subsite, T and N stage, chemotherapy, post-operative radiotherapy, whether teeth were extracted before radiotherapy, mandible volume, and the full dose-volume histogram (DVH) of the mandible: V5 to V70 Gy (percentage of the mandible receiving at least that dose), D2 to D99 (dose to a given percentage of the volume), and mean, minimum and maximum dose.

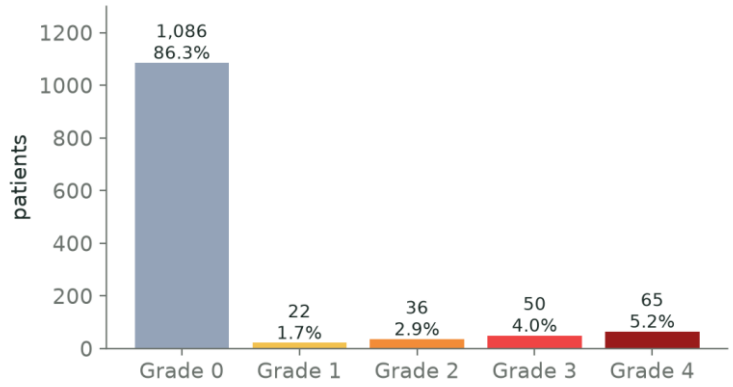


Figure 1. ORN grade in the cohort (n = 1,259). Grade 1 has only 22 patients.

Grade	Patients	Share	Grouping used
0 (no ORN)	1,086	86.3%	None
1	22	1.7%	Mild
2	36	2.9%	Mild
3	50	4.0%	Severe
4	65	5.2%	Severe

Table 2. Class distribution. ORN (grades 1 to 4) occurs in 173 patients, 13.7%.

Missing data are rare: only the chemotherapy column has gaps (20 patients), and these are kept as an "Unknown" category. The severe class imbalance is the central difficulty of the project: only 22 patients have grade 1 and 36 have grade 2.

3.2 What the exploration shows

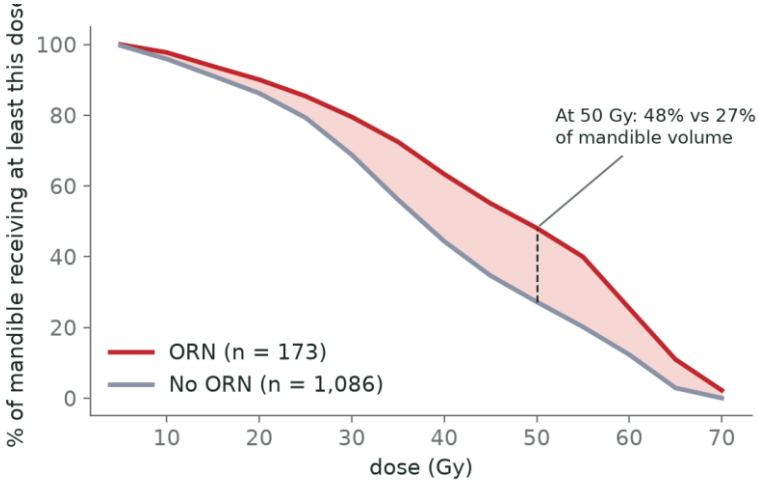


Figure 2. Median mandible DVH for patients with and without ORN. Patients with ORN have more of the mandible exposed at every dose from 20 Gy upward.

Dose separates the groups. At 50 Gy a median of 48% of the mandible is exposed in patients who developed ORN, against 27% in those who did not. The median D30 (the dose received by the hottest 30% of the mandible) is 58.5 Gy with ORN and 48.1 Gy without. Age, sex and mandible volume barely differ between the groups (median age 60 and 61 years; median mandible volume about 92,000 in both groups).

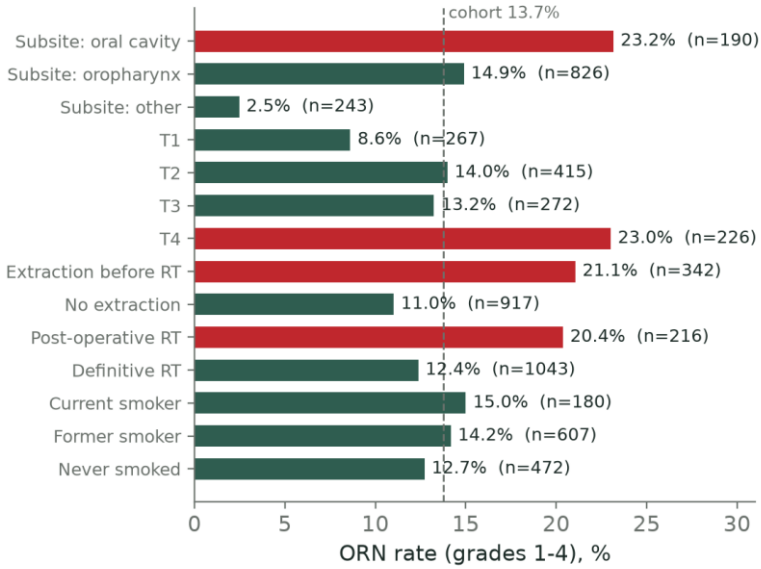


Figure 3. ORN rate by clinical factor (percentage of patients with ORN in each group; dashed line is the cohort rate of 13.7%).

Clinical factors matter too. Oral cavity tumours (23.2%) carry far more risk than the laryngeal, hypopharyngeal, nasopharyngeal and unknown-site group (2.5%). T4 tumours (23.0%) exceed T1 (8.6%). Patients who had teeth extracted before radiotherapy (21.1%) and those who had post-operative radiotherapy (20.4%) have higher rates. Smoking shows a small gradient (current 15.0%, former 14.2%, never 12.7%).

The DVH features are heavily redundant. Of the 780 pairs among the 40 dose columns, 224 have a Spearman correlation above 0.85 in absolute value. This is why we built a reduced feature set (set B, 16 features after pruning) as well as the full set (set A, 51 features) and a PCA version (set C).

4. Pipeline and modelling design

4.1 Pipeline

One command, `python backend/run_pipeline.py --data data.csv`, runs the whole chain. Its stages are:

- **Load and validate.** Schema, data type and range checks. Missing chemotherapy kept as "Unknown".
- **Preprocess.** A scikit-learn ColumnTransformer: one-hot for categories, ordinal coding for T and N stage, scaling fitted on the training data only.
- **Split.** Stratified 60 / 20 / 20 on the grade, seed 42, indices saved. About 755 training, 252 validation and 252 test patients.
- **Feature sets.** A: all 51 features. B: 16 after removing correlated DVH points. C: PCA on the DVH.
- **Cross-validate.** Five-fold cross-validation over the model grid, with SMOTE applied inside each fold only, so that synthetic samples never leak into validation.
- **Select and test.** The best configuration by macro-F1 (AUC for the binary task) is refitted on train plus validation. The test set is used once.

- **Evaluate.** 1,000-sample bootstrap confidence intervals, confusion matrices, ROC and precision-recall curves, quadratic weighted kappa (QWK).
- **Explain.** SHAP TreeExplainer (global and per-grade importance) and partial dependence.

Trained models and JSON result files are saved and served by a FastAPI backend (/predict, /explain, /results) to a Next.js research dashboard. The dashboard carries a disclaimer: research prototype, not validated for clinical use.

4.2 Modelling design

Dimension	Options
Target formulation	Binary (ORN yes / no); 3-class (none, mild = grades 1 to 2, severe = grades 3 to 4); two-stage (ORN? then mild vs severe); 5-class (grades 0 to 4)
Models	Logistic regression (L2), random forest, XGBoost, LightGBM, ordinal logistic regression (mord), stratified dummy baseline
Imbalance handling	None; class weights; SMOTE inside each fold

Table 3. The experimental grid.

Why not deep learning. With 1,259 tabular rows, tree models and regularised linear models are the standard choice, and Reber et al. found logistic regression beat a ResNet on this very dataset (F1 0.31 against 0.07). **Why macro-F1.** It weighs every grade equally, whereas accuracy would reward always predicting "no ORN" (86% accuracy for free). **Sanity check.** We replicated van Dijk's two-feature logistic regression (D30 plus dental extraction): cross-validated AUC 0.775 on our training split, in line with the published model.

The cross-validation results file records 44 runs (the slides say 47). Three of them, the binary XGBoost runs, failed with a configuration error (`num_class` set to 0) and carry no scores. Sets B and C were built but the recorded runs all use set A (43 runs) plus the van Dijk replication, so the comparison between feature sets that the slides imply was not actually run.

5. Results

5.1 Does ORN occur? (binary)

In five-fold cross-validation the binary models were close: random forest with class weights 0.755 AUC, LightGBM with SMOTE 0.752, logistic regression 0.747, against 0.48 for the dummy baseline. The random forest was selected and applied once to the 252 held-out patients.

Metric (test, n = 252, 34 with ORN)	Value	95% CI
ROC AUC	0.74	0.66 to 0.81
Average precision	0.32	baseline 0.13
Macro-F1	0.61	chance 0.48
Sensitivity / specificity at default threshold	0.29 / 0.91	0.14 to 0.45 / 0.87 to 0.95
PPV / NPV	0.34 / 0.89	
Brier score	0.127	
Sensitivity / specificity at threshold 0.25 (Youden)	0.76 / 0.61	optimistic: threshold chosen on the test set

Table 4. Binary random forest on the held-out test set.

The AUC of 0.74 equals the published van Dijk benchmark, which we treat as the pass mark for the pipeline. At the default threshold the model finds only 29% of ORN cases; moving the threshold to 0.25 raises sensitivity to 0.76 at the cost of specificity (0.61). The 0.25 threshold was found by maximising Youden's J on the test set itself (evaluate.py), so the 0.76 and 0.61 are optimistic and the "test set used once" rule was not fully kept for this one number. A fair version would choose the threshold on the validation split. Which operating point is right is also a clinical decision, not only a statistical one.

5.2 How severe? (3-class, two-stage, 5-class)

Formulation	Macro-F1 [95% CI]	Chance (dummy)	QWK
Binary	0.61	0.48	n/a
3-class	0.40 [0.33, 0.46]	0.36	0.22
Two-stage	0.42 [0.34, 0.49]	n/a	0.25
5-class	0.21 [0.18, 0.24]	0.20	0.17

Table 5. Test-set results by formulation. Chance is the stratified dummy's cross-validated macro-F1.

Grade	0	1	2	3	4
F1 (5-class model)	0.89	0.00	0.00	0.00	0.14
Test patients	218	4	7	10	13

Table 6. Per-grade F1 of the 5-class model. It predicts "no ORN" and almost nothing else.

The grade is not predictable yet. The 3-class model scores 0.40 against a chance level of 0.36, and its bootstrap interval is wide. The two-stage model is slightly better (0.42) because it first asks whether ORN occurs and only then asks about severity, but its interval overlaps the single-model result, so we cannot claim it is better. The five-class model collapses to grades 0 and 4 and scores 0.21, level with chance (0.20). Mild ORN (grades 1 and 2) is never predicted correctly in the 3-class or two-stage models. With only 4, 7, 10 and 13 test patients in grades 1 to 4, the per-grade numbers are also very noisy.

5.3 What drives the prediction

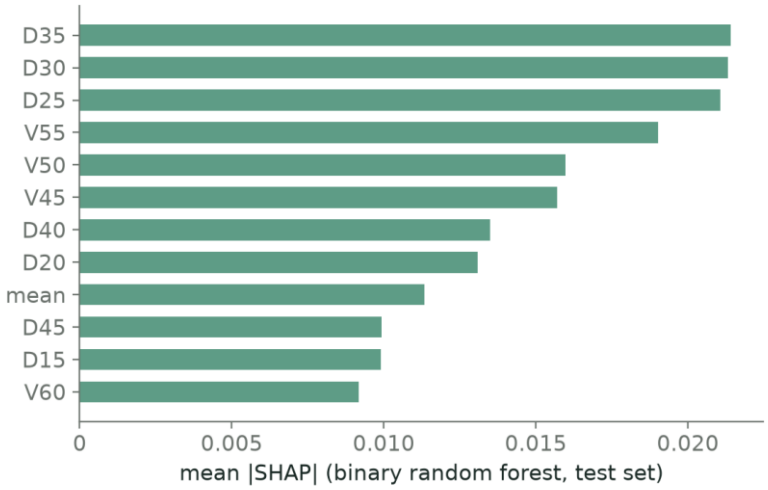


Figure 4. Twelve most important features for the binary random forest (mean absolute SHAP value on the test set).

Mid-range mandible dose dominates: D35, D30, D25, V55, V50, V45, D40 and D20 are the top eight. D30 is the same predictor that van Dijk and PREDMORN identified. Because the DVH points are strongly correlated, the model spreads credit among them, so the figure should be read as one "dose band" between about 25 and 55 Gy and not as eight independent effects. Dental extraction and clinical factors appear lower in the ranking.

5.4 Limitations of Part A

- **Few severe cases.** 22, 36, 50 and 65 patients in grades 1 to 4 limit what any model can learn; the test set holds only 34 patients with ORN.
- **One institution, one period.** All patients come from MD Anderson, treated 2005 to 2015. There is no external validation, and calibration was not studied.
- **Whole-mandible dose.** The DVH summarises a three-dimensional dose in one curve and cannot say where in the jaw the dose fell. This is the gap that Part B addresses.
- **Outcome definition.** ORN grade is the label recorded in the dataset; follow-up length and grading consistency are not available to us.
- **Experiments not run or not clean.** Feature sets B and C were not cross-validated, binary XGBoost failed (Section 4.2), and the tuned threshold was chosen on the test set (Section 5.1).

Part B. The IMPACT segmentation study

6. The IMPACT paper

Humbert-Vidan L, Castelo AH, He R, et al., for the OPC-SURVIVOR Program and the MD Anderson Head and Neck Cancer Symptom Working Group. *Image-based Mandibular and Maxillary Parcellation and Annotation using Computed Tomography (IMPACT).* medRxiv 2025, doi:10.1101/2025.03.18.25324199. It is a preprint and has not been peer reviewed.

6.1 Aim and cohort

The authors segment the jaw on radiotherapy planning CT into structures that match the ClinRad staging system. Sixty patients from the Stiefel Oropharynx Cancer Cohort were included (prescriptions of 60 Gy in 30 fractions to 70 Gy in 35 fractions; cases with more than 20 missing teeth or strong streak artefact were excluded). Fifty patients were used for training and ten for testing. The median age was about 59 years and the cohort was mostly male (47 of 50 in the training set). The mean number of missing teeth was 6 (range 0 to 18).

6.2 Structures and contouring

Twelve bone sub-volumes were contoured by hand in RayStation: left, right and central alveolar and basal regions of each of the mandible and the maxilla, with laterality set by grouping the teeth into three sextants. The alveolar region is an expansion from the alveolar crest of 5 mm inferiorly for the mandible and 3 mm superiorly for the maxilla. Teeth were contoured individually and numbered 1 to 32 in the ADA Universal system, with empty sockets of missing teeth included in the alveolar region. Overlaps were resolved with the central regions taking priority.

6.3 Method

Task	Structures	Model
Mandible	6 sub-volumes (+ background)	One stage: Swin UNETR on 96^3 patches
Maxilla	6 sub-volumes	Stage 1: ResUNet finds the whole maxilla region as one ROI. The image is cropped to 128^3 at the ROI centre of mass. Stage 2: Swin UNETR.
Teeth	32 teeth (+ background)	Same two-stage design with the ROI covering all teeth

Table 7. The three segmentation tasks (Section 2.4 of the paper).

Preprocessing: DICOM and RT structure sets converted to NIfTI (SimpleITK, dcmrtstruct2nii); resampling to 1 × 1 × 2 mm; Hounsfield units clipped to −200 to 1,600 and rescaled to 0 to 1. **Training:** MONAI, 500 epochs, learning rate 10^{-4}, Dice plus cross-entropy loss, augmentation by rotation (±12°), scaling (±10%) and intensity scaling and shifting (±10%), five-fold cross-validation on the 50 training scans. At test time the five fold models vote by majority. **Evaluation:** Dice, mean IoU, 95th-percentile Hausdorff distance (HD95) and average surface distance (ASD); then mean and maximum dose inside each predicted and manual structure, compared with a Wilcoxon signed-rank test and Bonferroni correction.

6.4 Reported results

Region	Dice	mIoU	HD (mm)	ASD (mm)
Mandible basal	0.85	0.74	1.17	0.30
Mandible alveolar	0.82	0.70	1.52	0.24
Maxilla alveolar	0.78	0.64	1.56	0.28
Upper central teeth	0.80	0.69	1.43	0.27
Upper premolars	0.69	0.56	2.30	0.62
Upper molars	0.76	0.64	3.65	0.57
Lower central teeth	0.76	0.63	1.62	0.34
Lower premolars	0.71	0.57	2.32	0.51
Lower molars	0.63	0.54	3.61	1.35

Table 8. Mean test-set metrics from Table 2 of the paper (10 test scans). The abstract gives lower premolars 0.70 and lower molars 0.71; we use the table. The maxilla basal sub-volumes were not segmented successfully and have no row.

Central teeth and mandible bone were segmented best; premolars and molars, which are often missing or distorted by fillings, worst. In the dose comparison most cases agreed within 2.5 Gy for mean dose, with larger outliers for maximum dose (up to 82% for molars). After Bonferroni correction only the central maxilla alveolar sub-volume differed significantly between manual and predicted contours (adjusted $p = 0.02$). Eight of the ten test cases were usable because two had no dose near the jaws.

Strengths of the paper are the clinical alignment of the structures, the open data and the dosimetric check in addition to overlap metrics. **Limits** the authors acknowledge or that follow from the design are: ten test scans, one institution, a preprint status, failure on maxilla basal bone, and weaker results for teeth that are often absent.

7. The public data and our exploration

7.1 What was released

File	Contents	Size
mandible.zip	60 CT scans + mandible masks (labels 1 to 6)	1.87 GB
maxilla.zip	the same 60 CTs + maxilla masks (labels 1 to 6)	1.87 GB
teeth.zip	the same 60 CTs + teeth masks (labels 1 to 32)	1.87 GB
dose_test_cases.zip	RT dose maps for 8 patients	4 MB

Table 9. figshare release, doi:10.6084/m9.figshare.28615874, licence CC BY 4.0.

Every zip repeats the same CT images, so the unique imaging is about 1.9 GB. The release has no train and test split file, no clinical table and no ORN outcome. All 180 masks lie on their CT's voxel grid (checked in the study deck). Labels 1 to 3 are basal and 4 to 6 alveolar; we inferred right, left and central from the position of each mask in the scan.

How we read the data. The 60 CTs were streamed from the mandible zip one at a time and never stored; only the masks, the dose maps and three example scans (101 MB) were kept. All statistics below come from the full set of 60.

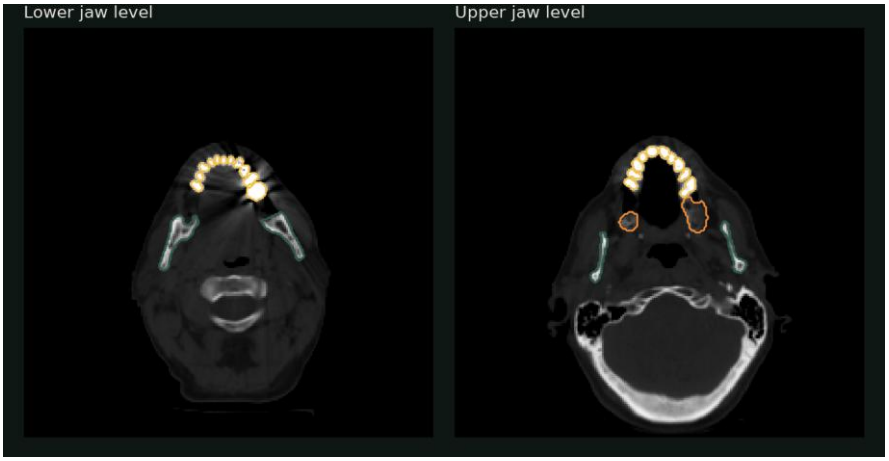


Figure 5. Patient 10, axial CT with masks: mandible (green), maxilla alveolar (orange), teeth (yellow). The lower slice shows streak artefact, likely from dental metal.

7.2 Scan characteristics

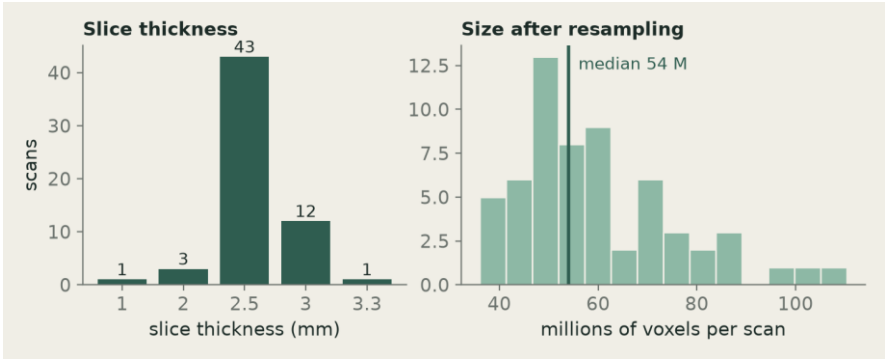


Figure 6. Slice thickness of the 60 scans (left) and scan size after resampling to 1 × 1 × 2 mm (right).

- **Matrix:** 512 × 512 in plane in all 60 scans, 101 to 296 slices (median 164), median scan length about 412 mm.
- **Pixel size:** 0.98 mm in 38 scans and 1.04 to 1.37 mm in the other 22, so fields of view run from 500 to 699 mm and resampling to 1 mm is needed.
- **Slice thickness:** 2.5 mm in 43 scans, 3.0 mm in 12, 2.0 mm in 3, and one each at 1.0 and 3.3 mm.
- **Hounsfield range:** most scans floor at −1,024 HU; four floor at −3,024 HU (padding values) and three at −1,000. All 60 contain voxels above the paper's 1,600 HU window, and 40 reach metal range (above 3,000 HU) in the study deck's count, so streak artefacts are present in the training data.
- **Size after the paper's resampling:** median 500 × 500 × 206 voxels (54 million), up to 699 × 699 × 285. After cropping to the body outline, as the authors' code does, the study deck found about 25 million voxels per scan (14 to 36 million).

7.3 Structures: maxilla basal is nearly empty

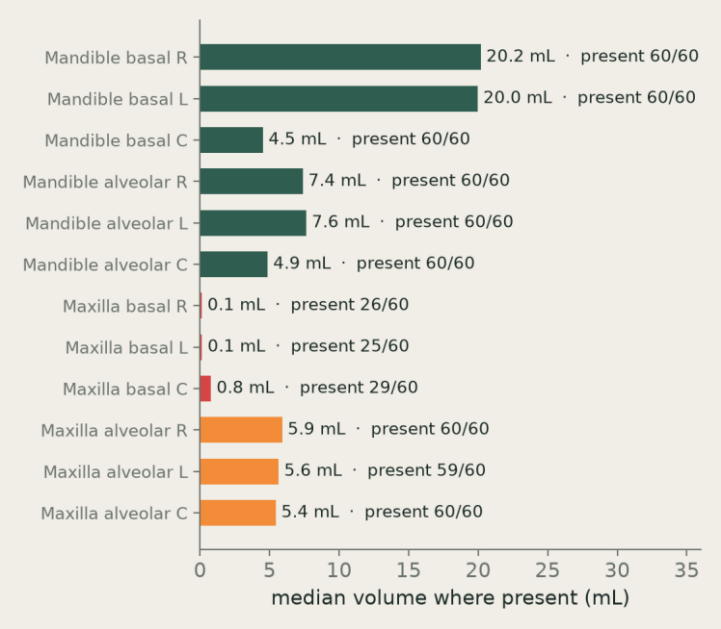


Figure 7. Median volume of each sub-volume where it is present, and the number of scans (of 60) in which it is present.

All six mandible sub-volumes and the three maxilla alveolar sub-volumes are present in 59 or 60 scans. The three maxilla **basal** sub-volumes are present in only 26 (right), 25 (left) and 29 (central) scans, and the lateral ones are 0.1 mL at the median, against about 20 mL for mandible basal bone. This is consistent with the paper's statement that the basal maxilla is often small or absent, and it means a network will have almost nothing to learn from. We expect any reproduction to fail on it too and recommend reporting it separately.

7.4 Missing teeth

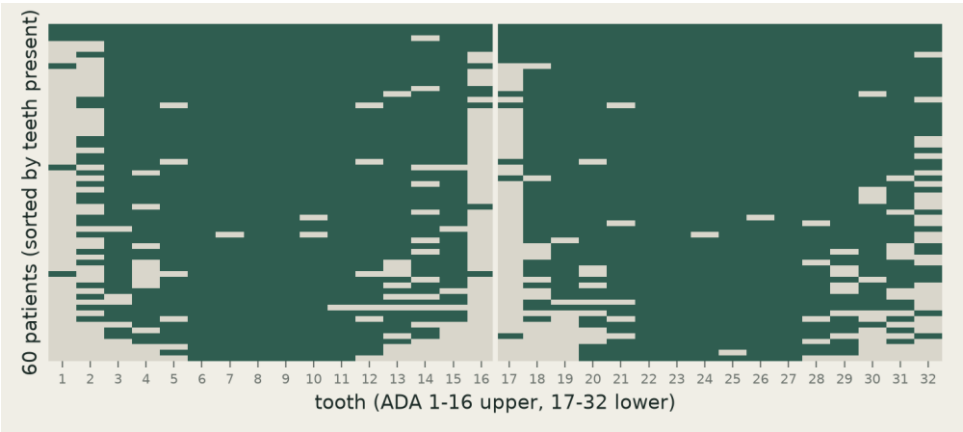


Figure 8. Presence of each tooth (ADA 1 to 32) in each of the 60 patients, sorted by number of teeth present. Light cells are missing teeth.

The median patient has 27 of 32 teeth (range 14 to 32), and 6.2 are missing on average, matching Table 1 of the paper. Third molars (teeth 1, 16, 17, 32) are present in only 25% of patients (6, 11, 12 and 32 scans). Among the other teeth the second molars and premolars go missing most often (tooth 2 is present in 23 scans). Presence is high for central teeth (99%) and lower for premolars (85% upper, 89% lower) and molars (52% upper, 64% lower).

Tooth group	Present	Paper Dice
Upper central	99%	0.80
Upper premolars	85%	0.69
Upper molars	52%	0.76
Lower central	99%	0.76
Lower premolars	89%	0.71
Lower molars	64%	0.63

Table 10. Share of tooth positions that are present (our count) against the paper's Dice (Table 2). Lower molars are both the least often present of the lower teeth and the lowest Dice. Upper molars do better than their presence suggests, so missing teeth are one factor and not the whole explanation.

7.5 Dose to the sub-volumes

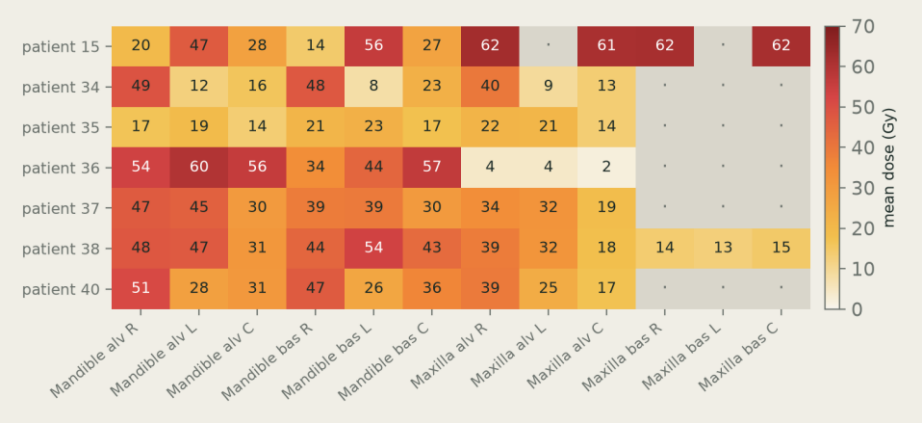


Figure 9. Mean dose (Gy) in each sub-volume for 7 of the 8 patients with dose maps, resampled onto the manual masks. Grey cells are structures absent from that patient.

The dose grids are 3 to 4 mm, coarser than the 1 × 1 × 2 mm masks, so each dose map was resampled onto the mask grid before averaging. Within one jaw the mean dose of a sub-volume can differ greatly from its neighbour: in patient 34 the left basal region receives 8 Gy and the right alveolar region 49 Gy. Between left and right basal mandible the gap reaches 42 Gy in patient 15 and 40 Gy in patient 34. A single whole-mandible mean would hide this, and it is the main argument for region-level dose.

Data issue. The dose file of patient 32 peaks at exactly 16,384 cGy (164 Gy), about twice the prescription, and the study deck reports a quarter of voxels above 80 Gy (we did not recheck that figure). We believe this is a scaling error and excluded the case, so the dose analysis rests on 7 patients: enough to illustrate the idea but not to test it.

8. Audit of the public code

The repository github.com/dr-mnaser/orodental-segmentation holds five files, one commit, and no README. We read every line of the five `seg_model.py` scripts: the mandible script, and a stage-1 (ROI) and a stage-2 (fine) script each for the maxilla and the teeth. They are near-copies; differences are in the number of classes, the ROI handling, the epoch count and some augmentation probabilities.

Script	Stage	Classes	Epochs
Mandible/seg_model.py	one stage	7	500
Maxilla/maxilla_roi	1, ROI	2	500
Maxilla/cropped_cm	2, fine	7	500
Teeth/teeth_roi	1, ROI	2	1000
Teeth/cropped_cm	2, fine	33	1000

Table 11. The five scripts.

Each script takes the fold number from the command line, uses 10 validation scans per fold, trains on four 96^3 patches per scan with the AdamW optimiser (learning rate 10^{-4}, weight decay 10^{-5}) and a Dice plus cross-entropy loss, validates with sliding-window inference every epoch, and saves the checkpoint with the best validation Dice. With 50 scans and 10 validation scans per fold, each fold trains on 40 scans.

Topic	Paper says	Public code does
Network	Swin UNETR	The SwinUNETR import is commented out. The default (use_unetr = False) trains a MONAI residual UNet; setting it to True builds a plain ViT UNETR.
Epochs	500	500 for mandible and maxilla; 1,000 in both teeth scripts.
Stage-2 crop	From the stage-1 output	Training and validation crops are centred on the ground-truth mask (GenerateROId on seg), so validation never sees stage-1 errors.
Test-time steps	Majority vote; Dice, mIoU, HD95, ASD; dose comparison	None of these are in the repository.
Data split	Random 50 / 10	Depends on the unsorted os.listdir order (seed 0).
Checkpoint choice	Not stated	test_size = 0: the best epoch is picked on the same fold it is scored on. ignore_empty = False gives an absent tooth with an empty prediction a Dice of 1, which flatters the score.
Augmentation	Rotation, scale, intensity	Rot90 with probability 0.5 in the mandible and maxilla stage-2 scripts and 0.1 elsewhere; random flips only in the two ROI scripts.
Packaging	n/a	Hard-coded cluster and Windows paths; maxilla_roi sets use_pretrained = True for a checkpoint that is not shipped. No README, requirements, weights, inference script or resume.

Table 12. Paper versus public code.

What this means. The repository shows the authors' settings but is a snapshot of the training step, not a pipeline. To reproduce the paper we must write the code that joins stage 1 to stage 2 at test time, the five-fold ensemble, the metrics and the dose comparison, and decide which network to use. The mismatch between paper and code is worth raising with the authors.

9. Can we train it? Cost and plan

9.1 Workload

There are five stage models (mandible, maxilla ROI, maxilla fine, teeth ROI, teeth fine) and five folds, so 25 training runs. Counting epochs as in the code gives 17,500 epochs (12,500 at the paper's 500 each). Each epoch has 40 scans × 4 patches = 160 patch passes, so training costs about 2.8 million patch passes, and validation adds roughly 3 million forward-only sliding windows, or about 5.8 million in the study deck's accounting.

Hardware	Seconds per patch	Time
Laptop CPU	1 to 3	67 to 200 days
Apple M-series GPU (needs code changes)	0.1 to 0.3	7 to 20 days
One data-centre GPU, e.g. A100	0.01 to 0.03	16 to 48 hours
GPU cluster, folds in parallel (authors)	n/a	hours per model

Table 13. Planning estimates from the study deck, from the workload counted in the scripts. They are not benchmarks.

These time estimates are uncertain by about a factor of ten. The figures above assume 10 to 30 ms per patch, which suits a small residual UNet (which is what the public scripts train by default). A Swin UNETR with the usual 48 feature channels is heavier; at an assumed 0.1 to 0.15 s per patch it would take on the order of 100 to 200 GPU-hours for the whole study. We have not measured either. The first task on a GPU should therefore be a short benchmark of one epoch of each network, and the plan below should be rescaled from that.

Other blockers: the scripts cache all training scans in memory (about 10 GB, a poor fit for an 18 GB laptop that also runs the operating system); no trained weights are published, so training cannot be skipped; and there is no resume, so a sleeping laptop or an expired free cloud session loses the run.

9.2 Our implementation

We wrote `impact_seg.py`, a single configurable MONAI script covering: scan listing, the paper's preprocessing and augmentation, ResUNet and Swin UNETR, five-fold splits, training with validation and best-checkpoint saving, two-stage prediction (stage-1 ROI, crop 128^3 at the centre of mass, stage-2 segmentation, majority vote across folds), and Dice, IoU, HD95 and ASD. It differs from the public code in using Swin UNETR as the paper states, cropping from the stage-1 prediction at inference, and ignoring classes absent from a validation scan instead of scoring them as 1.

Status. We tested it only on small synthetic volumes on a CPU: both training stages run for a few epochs and the prediction path runs end to end. Nothing has been trained on the real data and no Dice has been measured. Note that our fold split currently uses all 60 scans; to match the paper it should use the 50 training scans only, with the 10 test scans held out (the paper does not say which ten, so we would choose and document them).

9.3 Plan

Step	Work	Done when
1. Data and audit (done)	Release downloaded and explored, code audited	This report
2. Clean pipeline	Configurable code with a saved split, GPU or CPU, resume	A one-epoch run per network; benchmark times
3. Mandible first	One-stage model, fold 0 at reduced epochs, then all five folds	Basal Dice near 0.85 on validation
4. Two-stage models	Maxilla and teeth, stage 2 cropped from stage 1 at test time	End-to-end prediction on a held-out scan
5. Evaluate	Dice, mIoU, HD95, ASD against Table 8; dose comparison on the 7 usable cases	Table of our results next to the paper's

Table 14. Implementation plan. Steps 2 to 5 need GPU access (college server, TMH, or cloud credits).

Risks. Ten test scans give every Dice a wide uncertainty; maxilla basal bone and molars are rare in the data; the paper and the code disagree about the network and epochs; and compute is not yet secured.

10. Connecting the two parts

Part A shows that the whole-mandible DVH carries real signal for whether ORN occurs but little for how severe it is. Part B suggests why more detail might help: dose differs by tens of grays between neighbouring regions of one jaw, and the biology differs between alveolar and basal bone. A segmentation model of the IMPACT kind would let us compute, for each patient:

- mean, maximum and D30-style dose in each of the 12 sub-volumes (alveolar and basal, left, right and central, mandible and maxilla);
- the dose to individual teeth, relevant to extraction decisions;
- the fraction of each region above thresholds such as 50 and 60 Gy.

These would be added to, or replace, the DVH columns in the Part A pipeline, and the same evaluation (nested cross-validation, bootstrap intervals, SHAP) would test whether they improve on the baseline AUC of 0.74 or on grade prediction.

What this needs and what it cannot yet have. The public IMPACT release has no ORN outcomes or clinical data, so it cannot be joined to the Part A model. The route is a cohort with planning CT, dose and ORN outcomes for the same patients, for example the TMH cohort, with the segmentation model run on those scans. That requires the trained segmentation model (Section 9), data-sharing approval at TMH, and a check that a model trained on MD Anderson CT transfers to scans from a different scanner and protocol. We also have not studied MRI; our guide will share a paper on MRI-based segmentation, and the same plan would apply.

11. Conclusions

1. A tested, reproducible pipeline predicts whether ORN occurs at the level of the best published model (AUC 0.74), driven by mandibular dose in the 25 to 55 Gy band.
2. ORN grade cannot yet be predicted from dose and clinical data: three-class and five-class results are at or near chance, because severe grades are rare.
3. IMPACT offers a way to add spatial detail to the dose. Its public data are usable but small, lack outcomes, and are missing maxilla basal bone in about half the scans.
4. The public IMPACT code is incomplete and differs from the paper (no Swin UNETR, no inference, no ensemble, no metrics), so reproduction means building the pipeline, which we have started and tested on synthetic data.
5. The compute estimate for training is uncertain, from roughly one to several days on one GPU depending on the network, and must be measured first.

12. Future work

- Secure GPU access, benchmark both networks, and train the mandible model first.
- Run the cross-validation for feature sets B and C and fix the binary XGBoost configuration.

- Add calibration analysis and decision-curve analysis to Part A; choose an operating threshold with clinicians.
- Obtain a TMH cohort with CT, dose and ORN outcomes; run segmentation on it and test region-level dose features.
- Pool multi-centre data to grow grades 1 to 3, and validate externally.
- Read the MRI segmentation paper from our guide and compare it with the CT approach.

References

[1] Humbert-Vidan L, Castelo AH, He R, et al. Image-based Mandibular and Maxillary Parcellation and Annotation using Computed Tomography (IMPACT): a deep learning-based clinical tool for orodental dose estimation and osteoradionecrosis assessment. medRxiv 2025. doi:10.1101/2025.03.18.25324199.

[2] IMPACT data: figshare, doi:10.6084/m9.figshare.28615874. IMPACT code: github.com/dr-mnaser/orodental-segmentation.

[3] van Dijk LV, et al. Normal tissue complication probability (NTCP) prediction model for osteoradionecrosis of the mandible in patients with head and neck cancer after radiation therapy. Int J Radiat Oncol Biol Phys 2021. Dataset on figshare. (Our repository README cites Radiotherapy and Oncology; please confirm the journal.)

[4] Reber et al. 2023 (machine learning versus deep learning for ORN prediction) and PREDMORN 2025: cited as in our slides; full references to be confirmed.

[5] Hatamizadeh A, Nath V, Tang Y, et al. Swin UNETR: Swin transformers for semantic segmentation of brain tumors in MRI images. arXiv:2201.01266, 2022.

[6] Cardoso MJ, Li W, Brown R, et al. MONAI: an open-source framework for deep learning in healthcare. arXiv:2211.02701, 2022.

[7] Lundberg SM, Lee S-I. A unified approach to interpreting model predictions. NeurIPS 2017.

[8] Chawla NV, Bowyer KW, Hall LO, Kegelmeyer WP. SMOTE: synthetic minority over-sampling technique. J Artif Intell Res 2002;16:321-357.

[9] Chen T, Guestrin C. XGBoost: a scalable tree boosting system. KDD 2016.

[10] Ke G, Meng Q, Finley T, et al. LightGBM: a highly efficient gradient boosting decision tree. NeurIPS 2017.

Appendix A. Reproducibility

Part A. Clone github.com/MuaazSM/orn-pipeline. Backend: `pip install -r backend/requirements.txt`; train with `python backend/run_pipeline.py --data backend/data/raw/Final_final_used_data_Anonomyzed.csv`; serve with `uvicorn api.main:app --port 8000` and `npm run dev` in `frontend/`. Seeds are fixed (42). The random seed, split indices and trained models are saved in `backend/outputs/`.

Part B. The data come from figshare (four zips). Our scripts are in `impact-seg/`: `explore/fetch_stats.py` streams the CT statistics and saves the masks and dose maps; `explore/analyze_masks.py` computes volumes, presence and dose; `explore/make_figs.py` draws the figures; `impact_seg.py` is the segmentation code (`train` and `predict` commands); `make_synthetic.py` makes the synthetic volumes used for the smoke test. The training scripts expect `<case>/image.nii.gz` and `<case>/label.nii.gz`, whereas the release uses `image_<id>.nii.gz` and `rs_mask_<id>.nii.gz`, so a small renaming step is needed.

Appendix B. Discrepancies found while preparing this report

- The second deck ("IMPACT Implementation Study") is our own study of the MD Anderson paper, not a deck by the MD Anderson group.
- The IMPACT abstract gives lower premolars 0.70 and lower molars 0.71; its Table 2 gives 0.71 and 0.63. We use Table 2.
- The first deck says 47 configurations; the results file records 44 runs, of which 3 (binary XGBoost) failed.
- Feature sets B and C are described as built but all recorded cross-validation runs use set A.
- At the tuned threshold the saved sensitivity is 0.7647, which rounds to 0.76; the first deck says 0.77. This report uses 0.76. The threshold itself was chosen on the test set (Section 5.1).
- The repository README cites van Dijk 2021 in Radiotherapy and Oncology; the deck cites the Int J Radiat Oncol Biol Phys. Check against the paper.
- An earlier slide of ours counted 48 training scans per fold; with 50 scans and 10 validation scans per fold it is 40, which this report uses. The earlier 60 to 100 GPU-hour estimate has been replaced by the range in Section 9.1.
- The dose finding for patient 32 (a quarter of voxels above 80 Gy) comes from the study deck and was not rechecked here; the 16,384 cGy peak was.
- The paper's methods say Swin UNETR; the code builds UNETR or ResUNet (Section 8).

Appendix C. Glossary

Term	Meaning
ORN	Osteoradionecrosis: death of irradiated bone, graded 0 to 4 in this dataset
DVH	Dose-volume histogram: how much of an organ gets at least a given dose
Vx / Dx	Vx: percent of volume receiving at least x Gy. Dx: dose (Gy) received by the hottest x percent of the volume
Alveolar / basal	Bone around the teeth (spongy) versus the denser lower or upper body of the jaw
ClinRad	ORN staging system that grades the vertical extent of bone damage on imaging
Dice, mIoU	Overlap between predicted and manual masks, 0 (none) to 1 (perfect)
HD95, ASD	95th-percentile and average distance between predicted and manual surfaces, in mm
ROI	Region of interest
Swin UNETR / ResUNet	A transformer-based and a residual convolutional 3D segmentation network
Macro-F1	Average F1 over classes, each class counting equally
QWK	Quadratic weighted kappa: agreement that penalises distant grade errors more
SMOTE	Synthetic minority oversampling of rare classes
SHAP	Per-feature contributions to a model's prediction